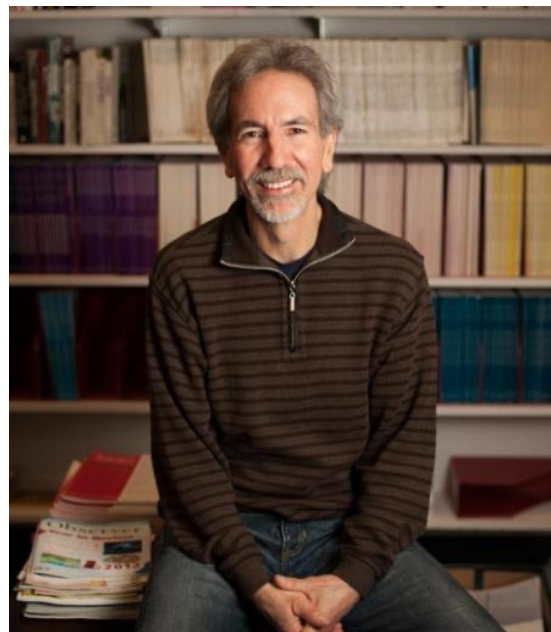


Individual Differences in Overconfidence

Jabin Binnendyk

Supervisor: Dr. Gordon Pennycook

Acknowledgments



Outline

What is overconfidence

How it is measured and potential confounds

A new measurement approach:

1- validation

2- task generalization

3 – predictive outcome generalization

Future directions

Overconfidence



Overconfidence

Fact check: It wasn't 'in jest.' Here are 53 times Trump said he'd end Ukraine war within 24 hours or before taking office



By Daniel Dale, CNN

🕒 16 min read · Published 5:07 PM EDT, Fri April 25, 2025



EMPIRICAL ARTICLE

Is overconfidence an individual difference?

Sophia Li¹, Randall Hale², and Don A. Moore¹

¹Haas School of Business, University of California Berkeley, Berkeley, CA, USA and ²Department of Psychology, UCLA, Los Angeles, CA, USA

Corresponding author: Sophia Li; Email: sophia-li@berkeley.edu

Received: 6 January 2025; **Revised:** 26 March 2025; **Accepted:** 28 March 2025

Keywords: overconfidence; individual differences; overestimation; overplacement; overprecision

Abstract

Some scholars have treated overconfidence as an individual difference—that is, assuming the tendency to be overconfident is stable within a person and differs meaningfully from person to person. We question this assumption. We investigate consistency within individuals between its three forms—overestimation, overplacement, and overprecision—in multiple domains (Study 1a and 1b), at multiple times (Study 1b and 2), and with multiple measures (Study 3a and 3b). We find mixed evidence of trait-like consistency. We do find some evidence of within-

The Trouble with Overconfidence

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Overconfidence: It Depends on How, What, and Whom You Ask

Joshua Klayman

University of Chicago

Jack B. Soll

INSEAD

Claudia González-Vallejo

Ohio University

and

Sema Barlas

Psychonomic Bulletin & Review
1997, 4 (3), 387–392

The domain specificity and generality of overconfidence: Individual differences in performance estimation bias

RICHARD F. WEST

James Madison University, Harrisonburg, Virginia

and

KEITH E. STANOVICH

University of Toronto, Toronto, Ontario, Canada

One hundred twenty-three college students performed a knowledge assessment task of motor skill in which they had to predict their performance before each block of trials. A bias in the direction of overconfidence on both tasks, even though the latter involved a main, did not require the use of numeric probabilities, and allowed predictions to be an aggregate judgment made in a frequentist mode. An analysis of individual differences that there was considerable domain specificity in confidence judgments. However, participants in showing overconfidence in the motor task, but not in the knowledge task.

Domain-specific account

Confidence is more malleable than we intuitively believe...

1) Feedback paradigms

Improves performance and calibration

2) Task difficulty

Hard-easy effect -> Inconsistent across tasks

3) Poor test-retest reliability

CONCLUSION:

There is an absence of evidence
supporting individual differences

Measurement

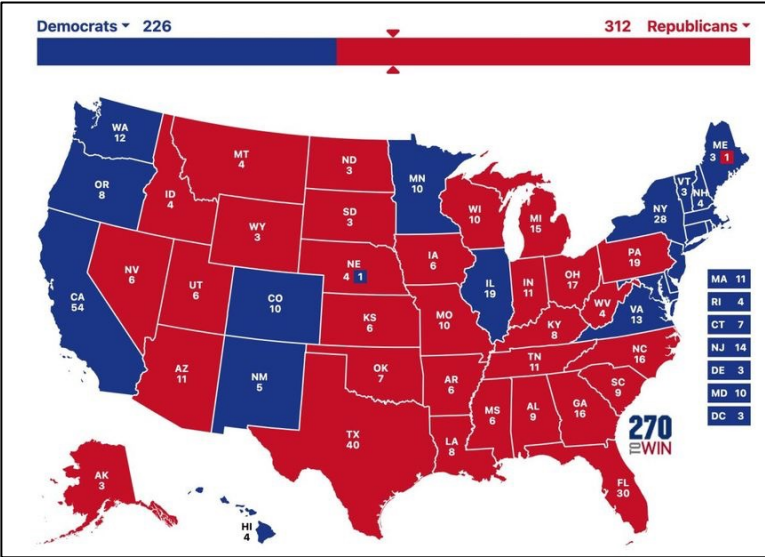
$$\text{Overconfidence} = \text{Confidence judgment} - \text{Accuracy}$$

*Other approaches exist but overestimation is the most common

Measurement



$$\begin{cases} 2x_1 + x_2 = 7 \\ x_1 + x_2 - 3x_3 = -10 \\ 6x_2 - 2x_3 + x_4 = 7 \\ 2x_3 - 3x_4 = 13 \end{cases}$$



Measurement

1. People vary in ability

Dunning-Kruger Effect: Low ability within a domain -> overestimate ability

Journal of Personality and Social Psychology
1999, Vol. 77, No. 6, 1121–1134

Copyright 1999 by the American Psychological Association, Inc.
0022-3514/99/\$3.00

Unskilled and Unaware of It: How Difficulties in Recognizing One's Own Incompetence Lead to Inflated Self-Assessments

Justin Kruger and David Dunning
Cornell University

People tend to hold overly favorable views of their abilities in many social and intellectual domains. The authors suggest that this overestimation occurs, in part, because people who are unskilled in these domains suffer a dual burden: Not only do these people reach erroneous conclusions and make unfortunate choices, but their incompetence robs them of the metacognitive ability to realize it. Across 4 studies, the authors found that participants scoring in the bottom quartile on tests of humor, grammar, and logic grossly overestimated their test performance and ability. Although their test scores put them in the 12th percentile, they estimated themselves to be in the 62nd. Several analyses linked this miscalibration to deficits in metacognitive skill, or the capacity to distinguish accuracy from error. Paradoxically, improving the skills of participants, and thus increasing their metacognitive competence, helped them recognize the limitations of their abilities.

Measurement

1. People vary in ability

Performance is an imperfect measure of ability

- Noise exists in performance measures
- Hinders ability to ‘fully control’ for ability

Widely-Used Measures of Overconfidence Are Confounded With Ability

76 Pages • Posted: 21 Jun 2023 • Last revised: 30 Nov 2024

[Stephen A. Spiller](#)

University of California, Los Angeles (UCLA) - Anderson School of Management

Date Written: November 29, 2024

Abstract

The overconfidence concept is one of the great success stories of psychological research, influencing research in other disciplines as well as discourse in the popular press, business, and public policy. Relative to underconfidence, overconfidence at various tasks is purportedly associated with greater narcissism, lower anxiety regarding those tasks, higher status, greater savings, more planning, and numerous other differences. Yet much of this evidence may merely reflect that there are associations with *ability* rather than *overconfidence*. This results from two underappreciated properties of typical measures of overconfidence. First, performance is an imperfect measure of ability; accounting for performance does not sufficiently account for ability. Second, self-evaluations of performance should reflect ability in addition to performance; because performance is ambiguous, people should use prior beliefs about their ability. I show these basic principles imply that commonly-used measures of overconfidence are confounded with ability. I support these analytical results by reexamining previously-published findings. In the first analysis, I find overconfidence predicts subsequent performance, consistent with overconfidence as a signal of ability but inconsistent with overconfidence as a bias. In the second set of analyses, I find the purported association between overconfidence and other proposed constructs can be adequately explained through ability alone. I close with recommendations on approaches to recognize and reduce the extent of the problem. This model serves as a stark reminder: when researchers propose that differences in overconfidence are associated with other behaviors, beliefs, or evaluations, they must account for the possibility that differences in ability provide a sufficient explanation.

Keywords: overconfidence, ability, knowledge, performance, measurement error

Measurement

2. *a priori* beliefs

Task familiarity cues expectations

- could bias estimated performance
- especially when performance is ambiguous.

Widely-Used Measures of Overconfidence Are Confounded With Ability

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Keywords: overconfidence, ability, knowledge, performance, measurement error

Measurement

$$\text{Overconfidence} = \text{Confidence judgment} - \text{Accuracy}$$

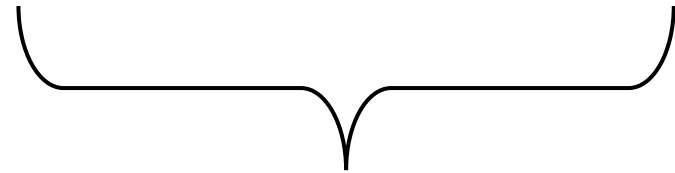


a priori beliefs



Vary in ability

Imperfect measure



Potential Confounds

IMPLICATION:

Task confounds hinder ability to
detect individual differences

A new approach

Create a new task that:

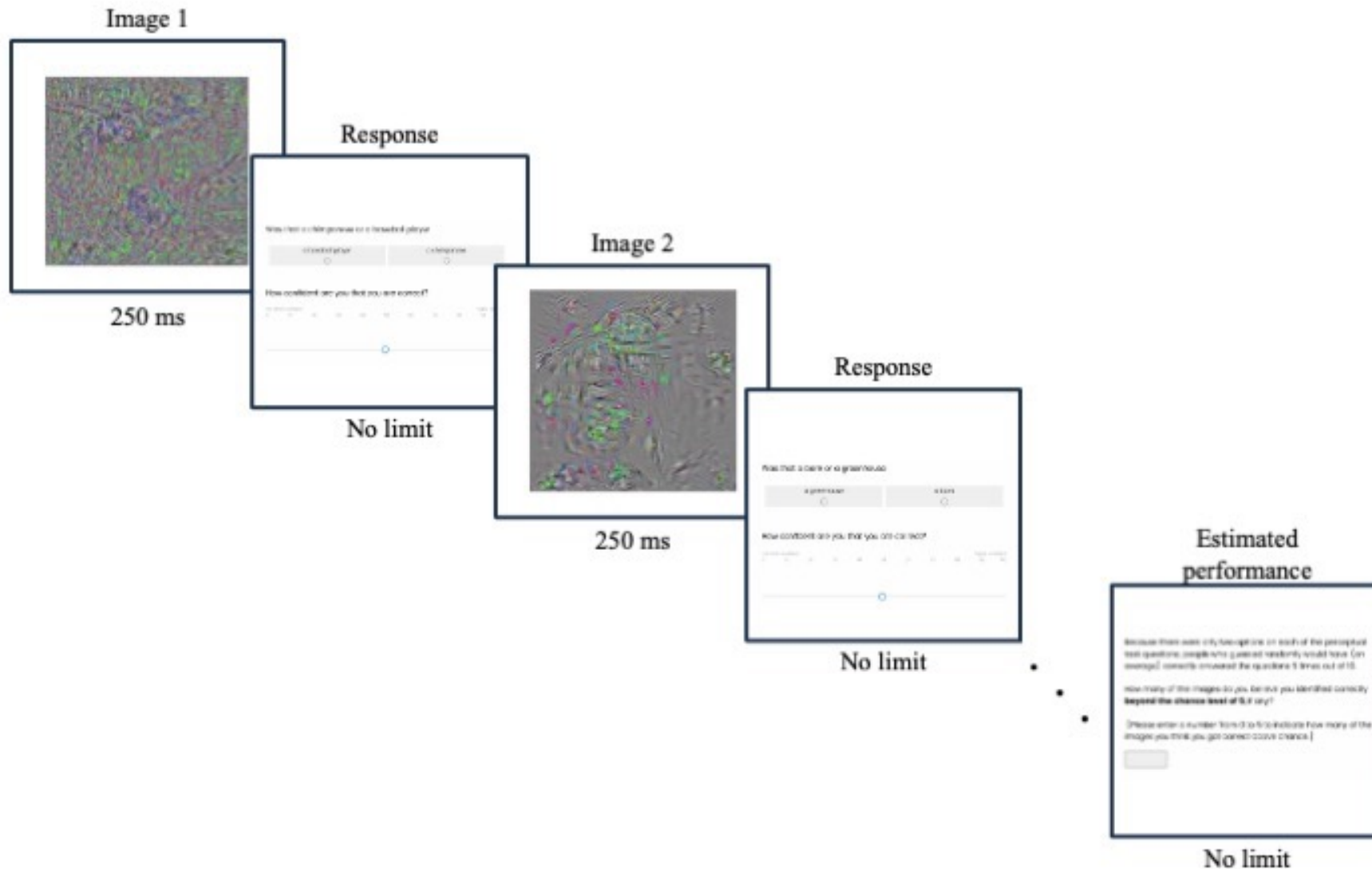
1. Performance rarely exceeds chance
2. Is not skill based
3. Is novel

Generalized Overconfidence Task (GOT)

In this section of the study, you will be presented with scrambled images very rapidly. Please do your best to identify what is in the scrambled images.



Generalized Overconfidence Task (GOT)



1 - validation

1 - validation



- 4 studies (N = 1,293)
 - Predictive validity (Study 1a)
 - Divergent validity (Study 1b)
 - Test-retest reliability (Study 2a / 2b)
- * Contrasted GOT to a task-specific approach (cognitive and numeracy tasks)

Judgment and Decision Making (2024), vol. 19:e28 1–24
[doi:10.1017/jdm.2024.22](https://doi.org/10.1017/jdm.2024.22)



EMPIRICAL ARTICLE

Individual differences in overconfidence: A new measurement approach

Jabin Binnendyk^{1,2} and Gordon Pennycook^{1,2,3}

¹Department of Psychology, Cornell University, Ithaca, NY, USA; ²Department of Psychology, University of Regina, Regina, SK, Canada and ³Hill/Levene Schools of Business, University of Regina, Regina, SK, Canada

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Received: 16 January 2024; **Revised:** 17 June 2024; **Accepted:** 17 June 2024

Keywords: overconfidence; general overconfidence; scale validation; calibration

1 - validation

- Predictive validity (Study 1a)



1 - validation



- Predictive validity (Study 1a)
 - Strong predictor of outcomes previously associated with overconfidence
 - Conspiracies ($r = .34$)
 - Overclaiming ($r = .30$)
 - Bullshit receptivity ($r = .34$)
 - Fake news headlines ($r = -.28$)

1 - validation

- Divergent validity (Study 1b)





1 - validation

- Divergent validity (Study 1b)
 - Thinking Styles ($r_s = .02 - .24$)
 - Big 5 Personality Inventory ($r_s = .07 - .16$)
 - Narcissism ($r = .16$)
 - Dogmatism ($r = .26$)
 - Optimism ($r = .08$)
 - Self-efficacy ($r = .16$)
 - Wishful thinking ($r = .10$)
 - Illusion of control ($r = .03$)



1 - validation

- Test-retest reliability (Study 2a / 2b)
 - Across 15-days:
 - GOT Estimate: $r = .54$ and $.53$; Confidence sliders: $r = .77$
 - Numeracy Overconfidence: $r = .18$ and $.37$
 - Cognitive Overconfidence: $r = .36$ and $.42$

1 - summary

- GOT displays:
 - Face, predictive, and divergent validity
- Addressing confounds allowed a general overconfidence to emerge
 - Test-retest reliability

16 October 1964, Volume 146, Number 3642

SCIENCE

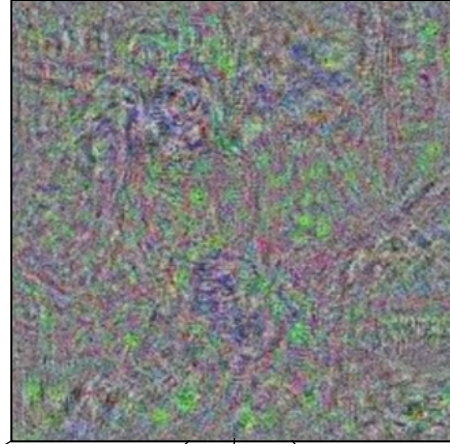
Strong Inference

Certain systematic methods of scientific thinking
may produce much more rapid progress than others.

John R. Platt



Finding



$H1_1$

$H1_2$

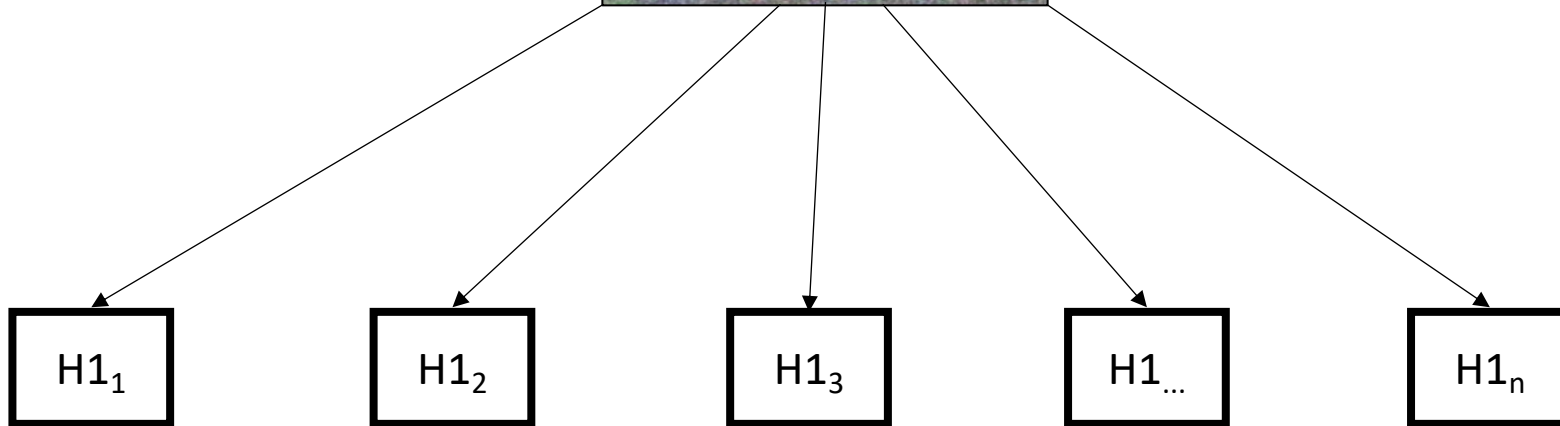
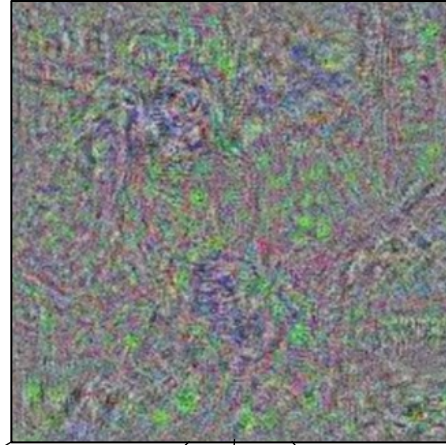
$H1_3$

$H1_{\dots}$

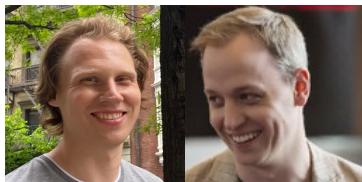
$H1_n$

Possible Explanations

Finding



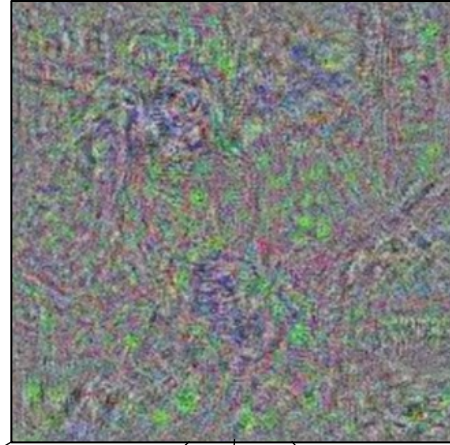
...



Possible Explanations

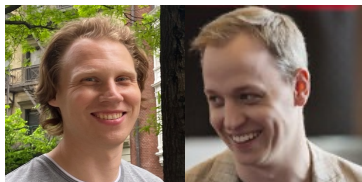
“Obvs proves OC is a trait”

Finding



$H1_1$

⋮



$H1_2$

⋮

Task
specific?

$H1_3$

⋮

Not predictive
of real-world
outcomes?

$H1_{...}$

⋮

Another
explanation

$H1_n$

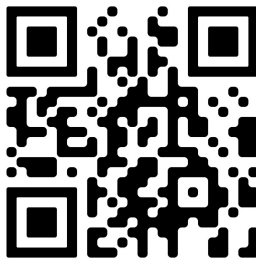
⋮

And others

“Obvs proves OC is a trait”

2 – task generalization:
An adversarial collaboration

2 – task generalization



Psychological Science
Volume 36, Issue 12, December 2025, Pages 941-951
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<https://doi.org/10.1177/09567976251401538>

Sage Journals

General Article - Research Article

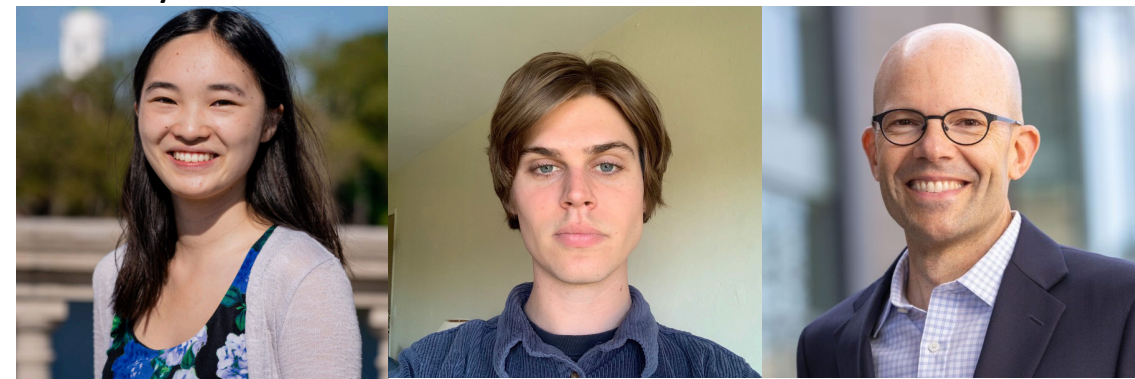
Is Overconfidence a Trait? An Adversarial Collaboration

Jabin Binnendyk ¹, Sophia Li ², Thomas Costello ³, Randall Hale⁴, Don A. Moore², and Gordon Pennycook ^{1,5}

Cornell Team:



Berkeley Team:

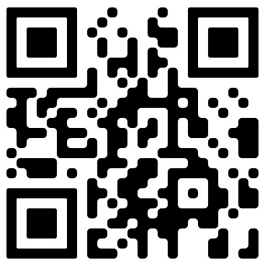


A new approach *revisited*

Create a new *tasks* that:

1. Performance rarely exceeds chance
2. Is not skill based
3. Is novel

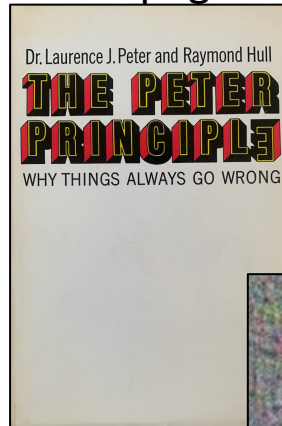
2 - task creation



Sibling order



Book pages



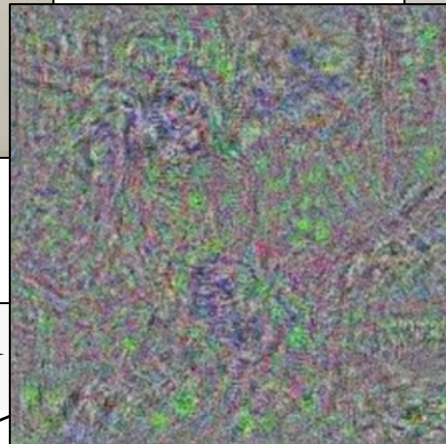
Birthday day



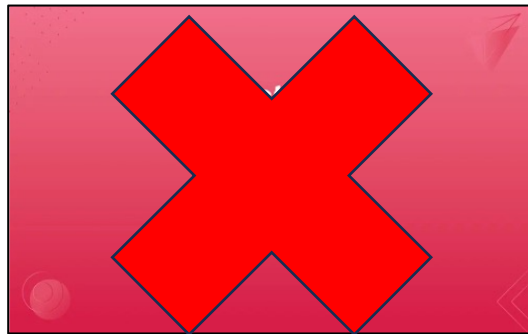
Bellybutton



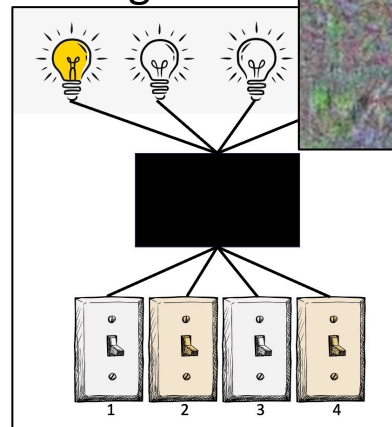
GOT



Preferences



Lightbulb



Card suit



Age (32 or 33)



Criterion to declare victory

rs $\leq$.3

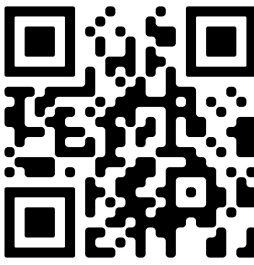


rs .3 - .4



rs $\geq$.4





2 - results

2 key findings:

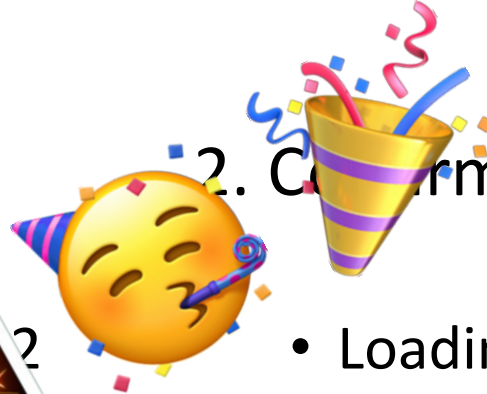
1. Correlation across tasks:

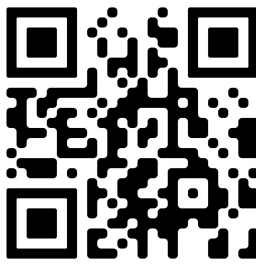
- Average confidence: $r = .42$
- Estimated performance: $r = .42$



2. Confirmatory Factor Analysis:

- Loadings: .72 - .88
- Excellent CFI, TLI, SRMR, and adequate RMSEA





2 - results: CFA

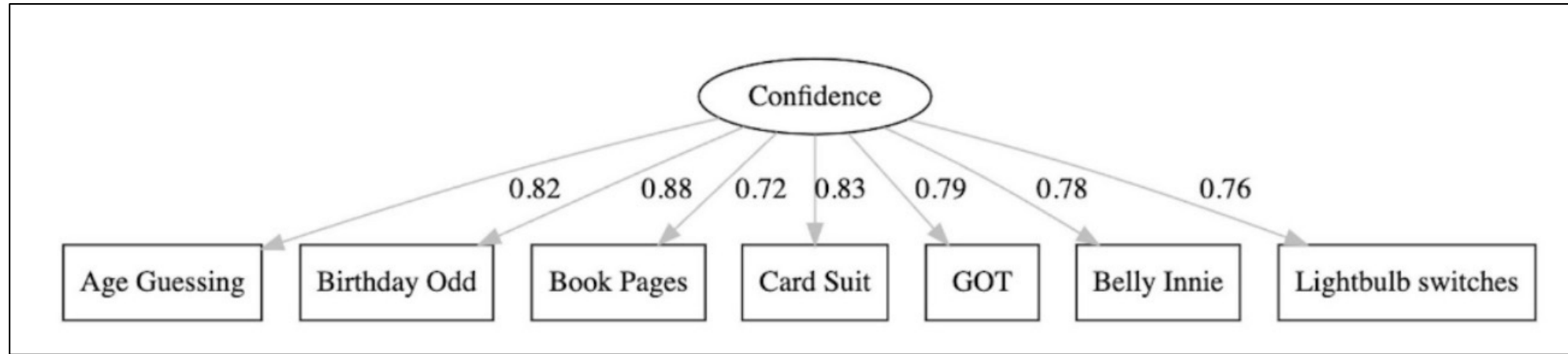


Table 3. *Fit statistics for a one-factor CFA using average task confidence scores.*

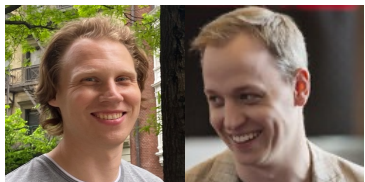
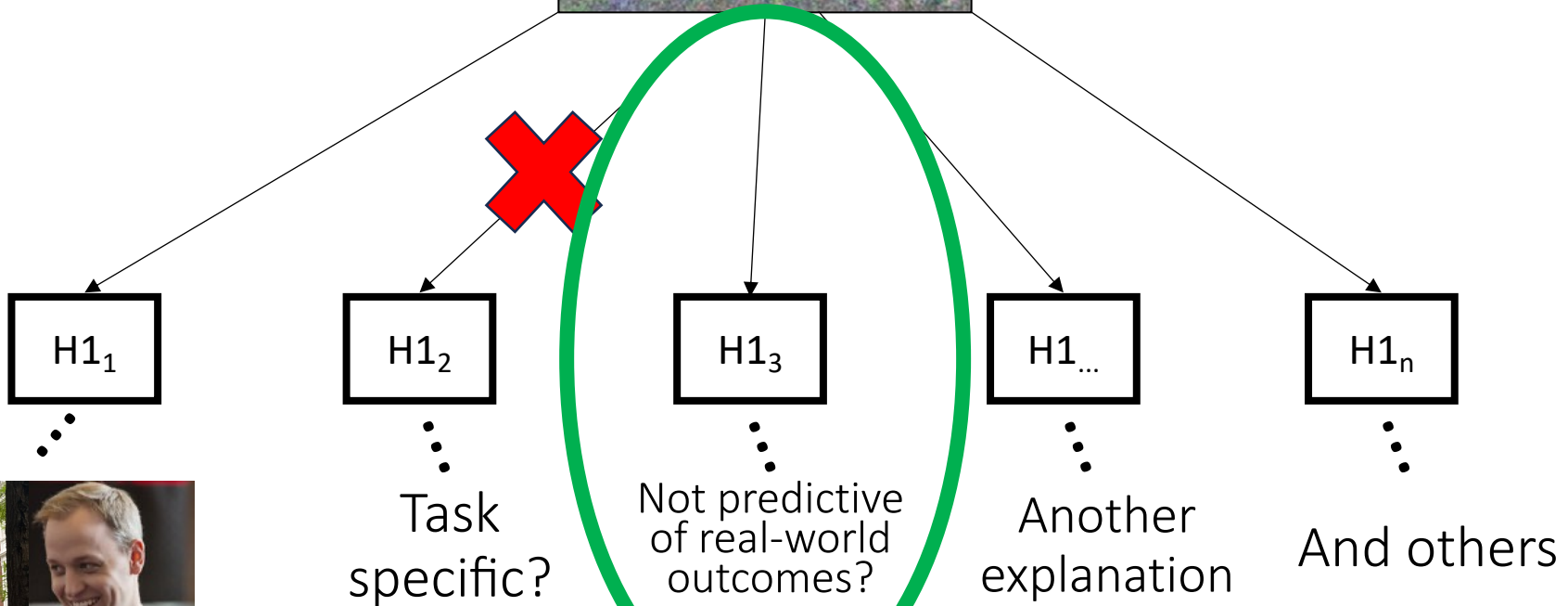
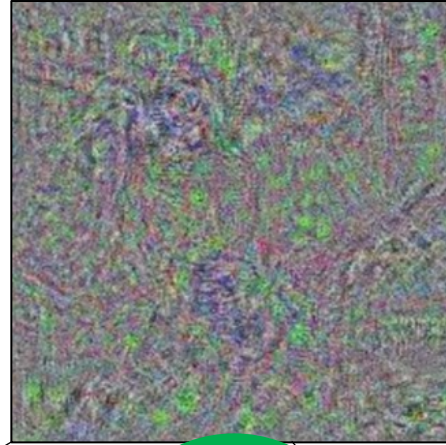
χ^2	df	χ^2/df	p	CFI	TLI	RMSEA [90% CI]	SRMR	AIC	BIC
56.21	14	4.01	< .001	.98	.97	.06 [.04, .07]	.03	28,997.42	29,099.72

2 - summary

- GOT style tasks suggest an underlying trait
 - 'Nonsense' tasks were highly correlated
- CFA findings provide additional support
 - Strong loadings
 - Data best explained with a one factor solution



Finding



“Obvs proves OC is a trait”

3 – predictive outcome generalization

3 – predictive outcome generalization

- 3 studies – current set of projects
 - S1 - Endorsing implausible statements
 - S2 – Updating sufficiently
 - S3 – Trust in experts

3 – predictive outcome generalization

How do we know it predicts the type of overconfidence we care about

and is not limited to your ‘toy tasks’?

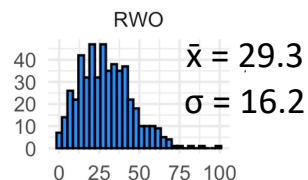
... “Let’s ask them about a variety of situations” ...

3 – predictive outcome generalization

S1: Does the GOT predict endorsing implausible statements?

Created 75 items ranging in plausibility: participants rated 10

Least plausible*



Most plausible



I could outrun a tornado*

(6.69)



I could represent myself in
a criminal trial and get the
charges dismissed (20.78)



If asked to produce a paper
or report I could finish it by
tomorrow (69.16)

3 – predictive outcome generalization

Additional measures:

Actively open-minded thinking -> “Whether something feels true is more important than evidence.”

AOT- SJT -> Scenarios where ppl choose between 3 options– captures situational AOT

Overclaiming -> How familiar are you with ... Some items are made-up

Intellectual humility -> “I reconsider my opinions when presented with new evidence.”

3 – predictive outcome generalization

S1 Results

Correlations with statements

Regression: $\text{lm}(\text{items} \sim \text{GOT} + \text{AOT} + \text{AOT-SJT} + \text{OC})$

GOT -> $r = .39, p < .001$ -----> $\beta = .34, p < .001$

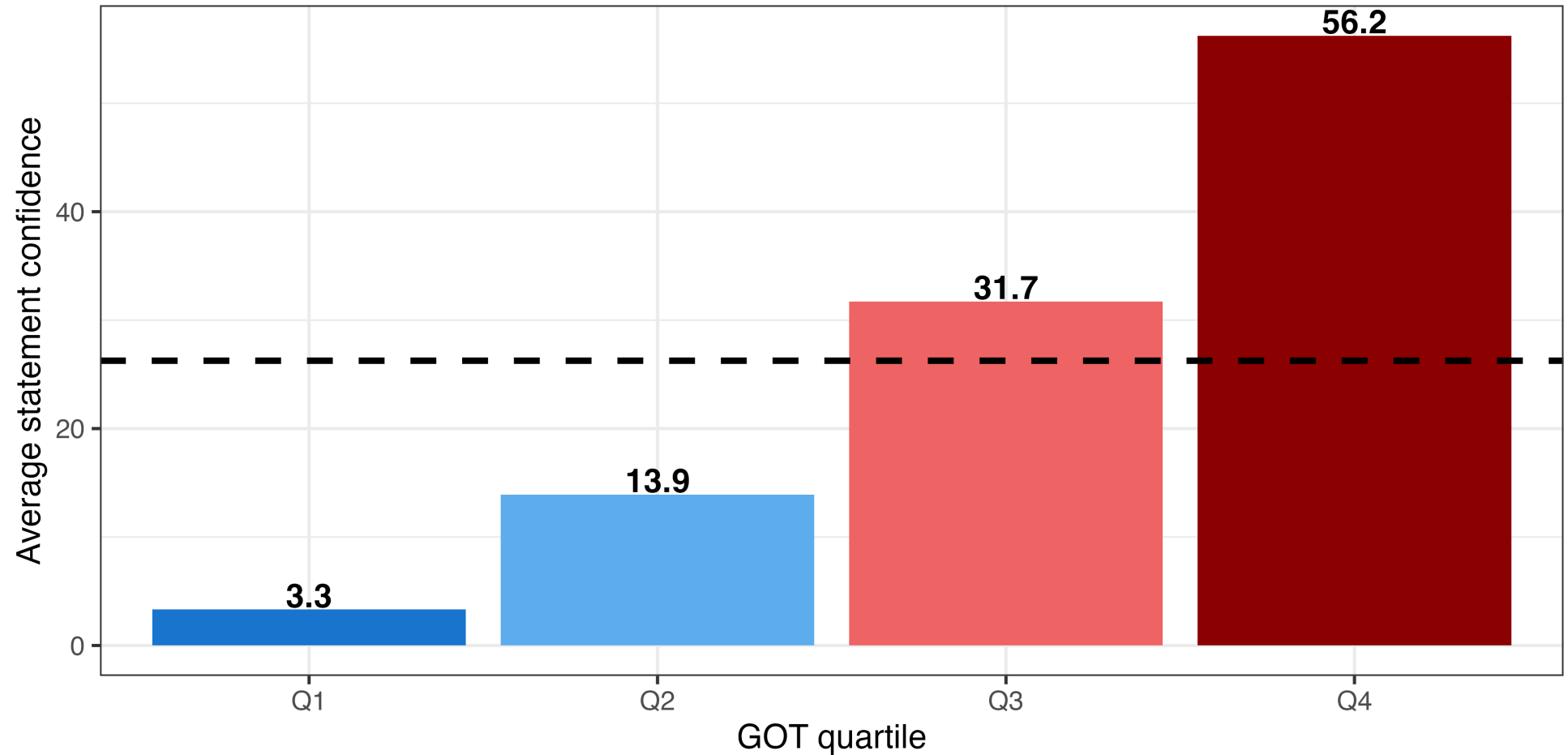
Actively open-minded thinking -> $r = -.29, p < .001$ -----> $\beta = -.20, p < .001$

AOT- SJT -> $r = -.10, p = .021$ -----> $\beta = -.04, p = .365$

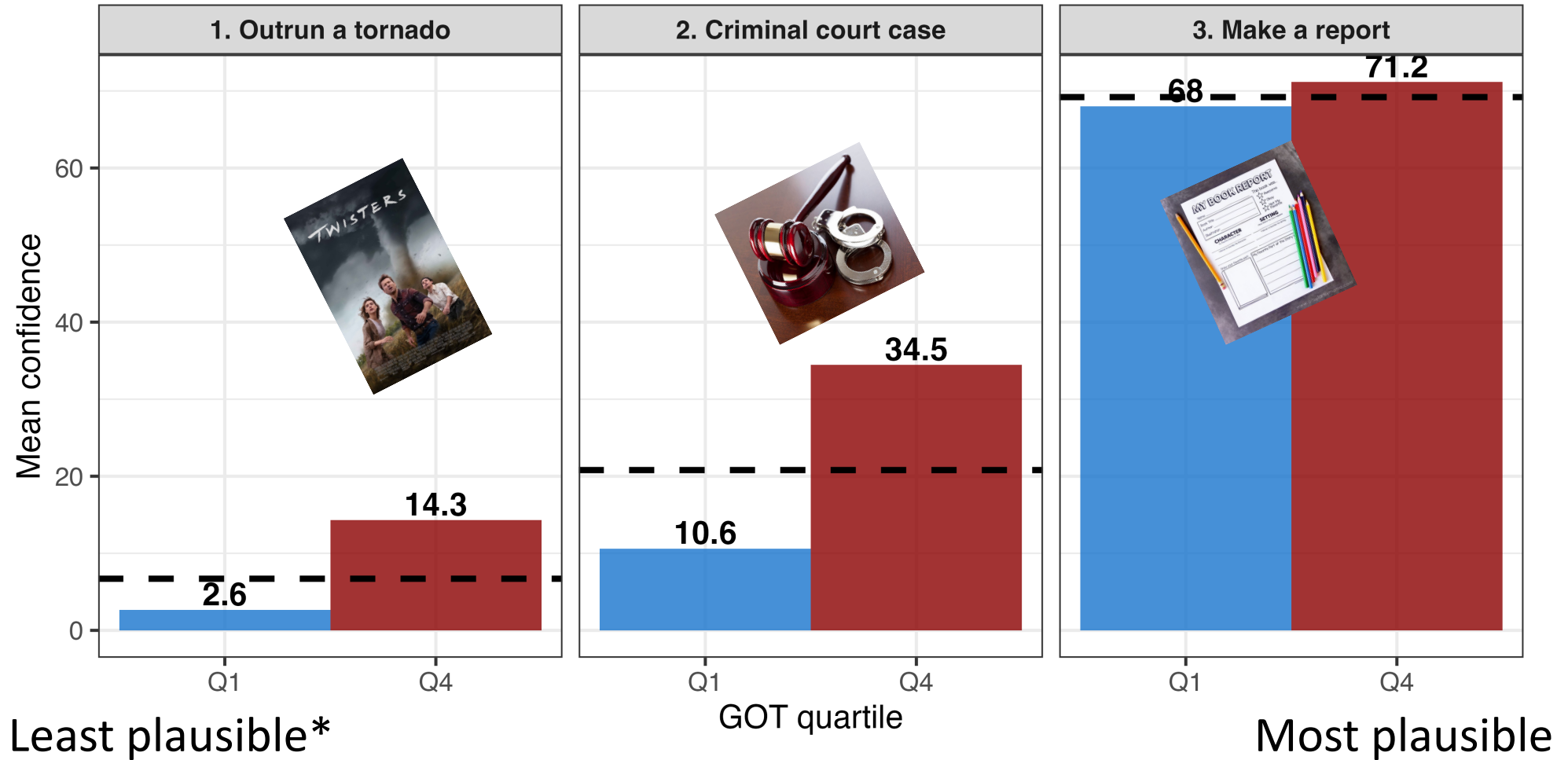
Overclaiming -> $r = .31, p < .001$ -----> $\beta = .03, p = .423$

Intellectual humility -> $r = -.01, p = \text{n.s.}$

3 – predictive outcome generalization

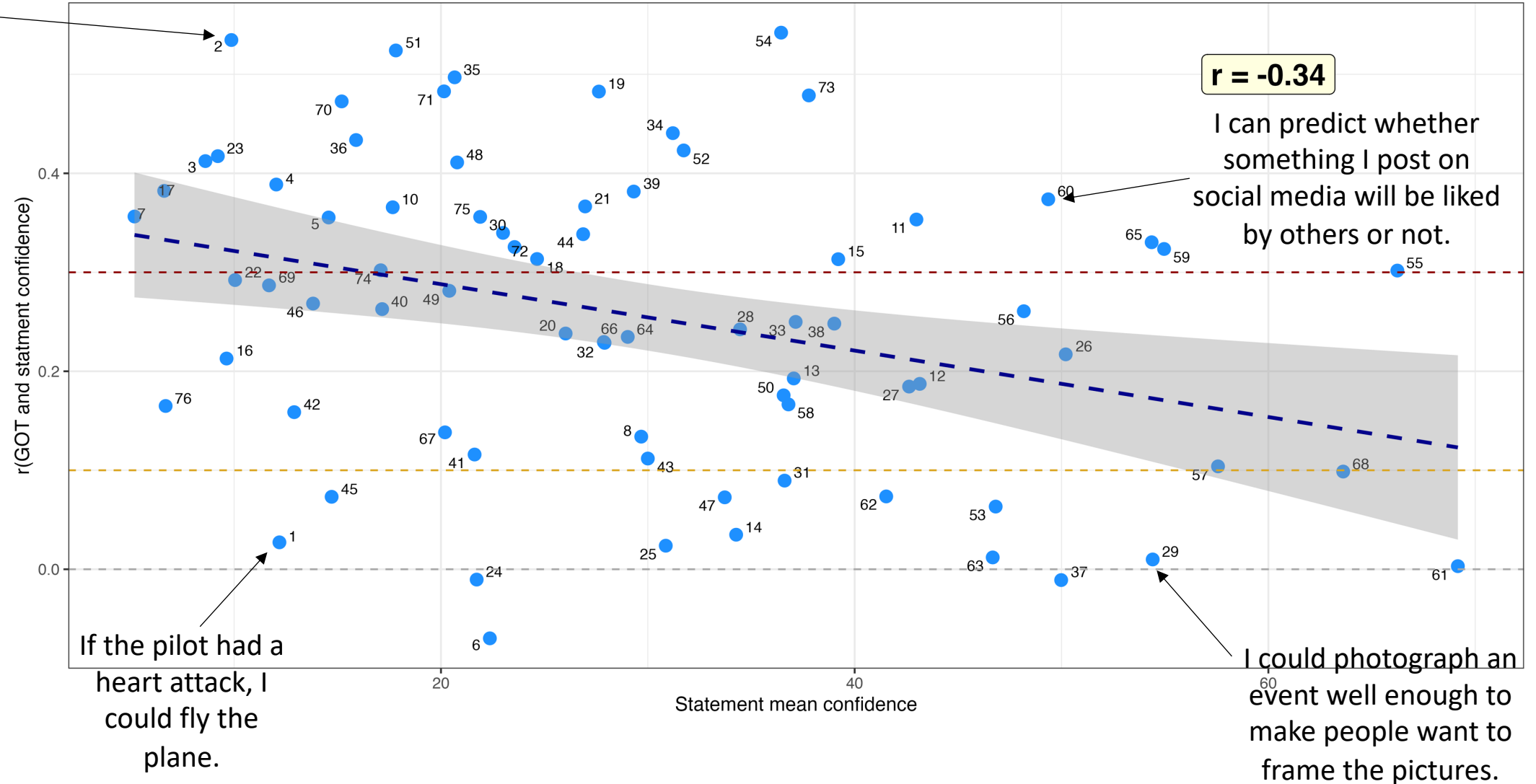


3 – predictive outcome generalization



3 – predictive outcome generalization

I could score a point on Serena Williams, a professional women's tennis player, in a tennis match.



3 – summary

- GOT confidence -> greater confidence for statements
 - Even when accounting for other individual differences
- Item analysis shows overconfident ppl were more confident for *less* plausible statements.

3 – predictive outcome generalization

How do we know it predicts the type of overconfidence we care about

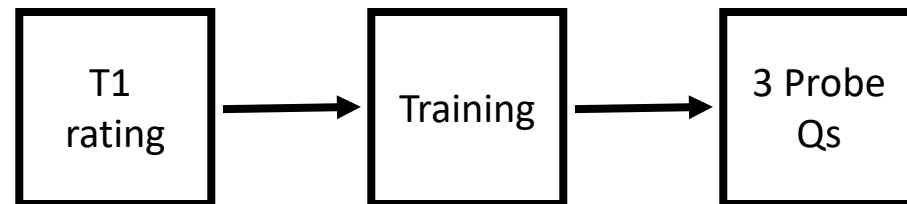
and is not limited to your ‘toy tasks’?

... Do overconfident people update sufficiently? ...

3 – predictive outcome generalization

S2: GOT -> Illusion of Explanatory Depth paradigm (IOED)

Idea is to expose gaps in one's knowledge...



Additional measures: Actively open-minded thinking, Intellectual Humility, and Overclaiming

Please describe, in as much detail as possible, how a zipper works.

What ensures that once the teeth separate under tension?

T2 rating

Level 7 Knowledge: A person with level 7 knowledge has very deep and detailed knowledge of how a cross bow works. For instance they would know it has a stiff, flexible piece of metal as a bow with a wire or strong line; that the bow is permanently mounted on a block of wood or metal; that the wire is pulled back by something that gives a mechanical advantage, either a lever, or small block and tackle, or by a crank wound around a spool that pulls a wire attached to the bow wire. The bow wire is then held back by a pin that is connected to a trigger, and an arrow is set in front of it. Often the pin is forked so the arrow can sit directly in the wire. The pin is directly connected to the trigger so that when you pull on the trigger, it causes it to pivot around a point such that the end that is the pin moves downwards and releases the bow wire. When the pin releases the string, the bow very quickly un-flexes, rapidly imparting all the energy stored in the flexed bow to the arrow. If this person were to draw a diagram of a crossbow it might look like this.

The diagram shows a crossbow with an arrow. Labels include: 'Bow permanently fixed to cross piece' pointing to the bow, 'Bow releases stored energy very fast when trigger is pressed' pointing to the trigger mechanism, 'Trigger Catch' pointing to the trigger, 'Strong Cord' pointing to the cord, 'Device for drawing back bow gives mechanical advantage' pointing to the handle, 'Strong Bow' pointing to the bow itself, and 'Bolt leaves with lots of energy' pointing to the arrow.

3 – predictive outcome generalization

S2: GOT -> Illusion of Explanatory Depth paradigm (IOED)

Results: Correlations difference score (t1 – t2 rating) Regression: $\text{lm}(t2 \sim t1 + \text{GOT} + \text{AOT} + \text{OC} + \text{IH})$

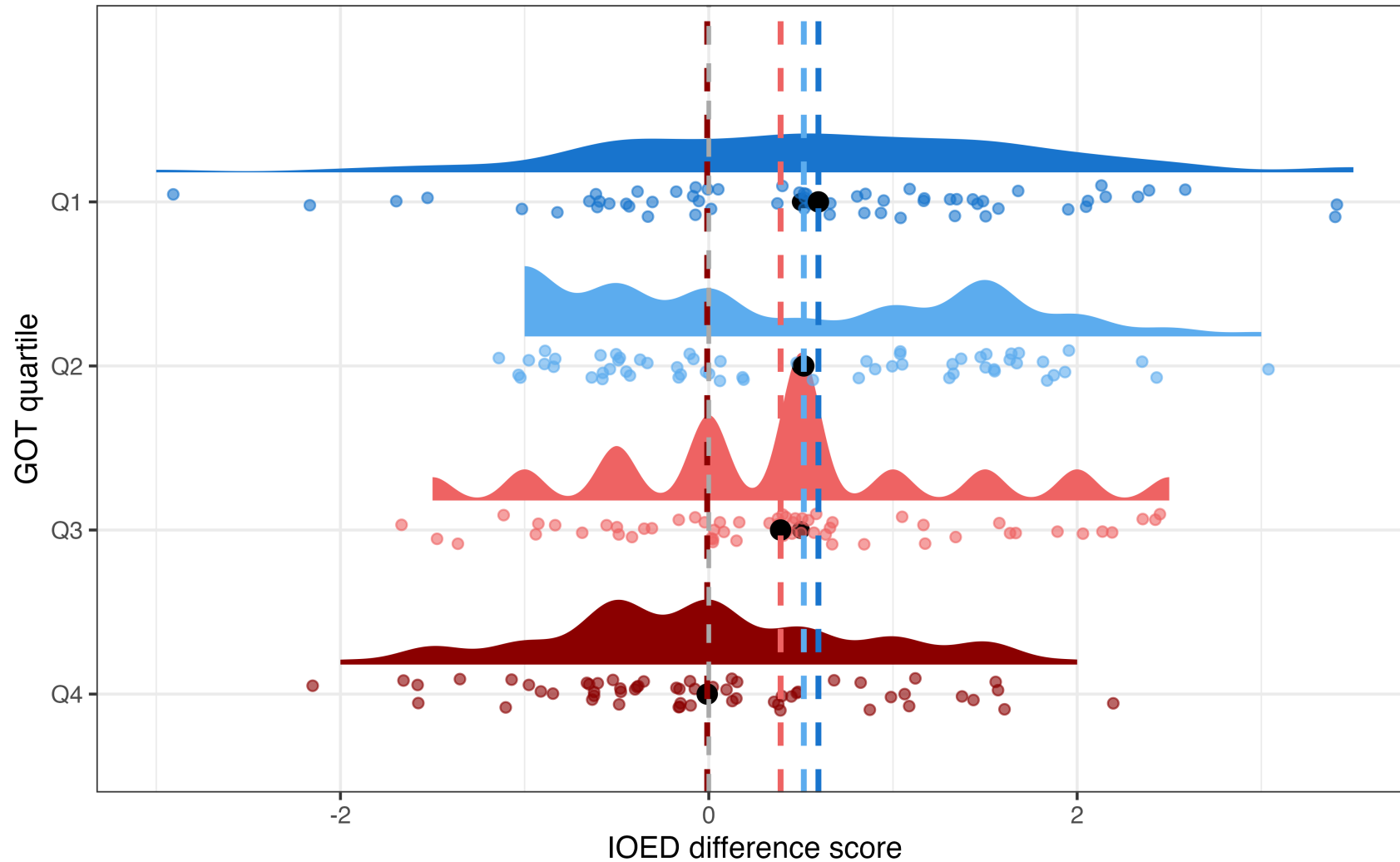
GOT -> $r = -.20, p = .002$ -----> $\beta = .19, p < .001$

Actively open-minded thinking -> $r = .08, p = .221$ -----> $\beta = -.07, p = .156$

Overclaiming -> $r = -.01, p = .857$ -----> $\beta = .01, p = .817$

Intellectual humility -> $r = .01, p = .823$ -----> $\beta = .05, p = .287$

3 – predictive outcome generalization



3 – summary

- Greater confidence on GOT -> less updating after gaps exposed
 - Highest GOT quartile doesn't show intervention effect
- GOT accounts for the most variance, even when other individual measures are included.

3 – predictive outcome generalization

How do we know it predicts the type of overconfidence we care about

and is not limited to your ‘toy tasks’?

... Trust in experts? ...

3 – predictive outcome generalization



3 – predictive outcome generalization

We are interested in how much people know about a variety of different topics.

Please indicate, on a scale from 0 to 100, how much you know about each topic.

0 = You know nothing at all about the topic

100 = You know all there is to know about the topic

0 10 20 30 40 50 60 70 80 90 100

The physical, chemical, and biological properties of food



Bees and honey farming



Astrophysics



History



We are interested in how much people think that experts know about a variety of different topics. That is, if you were in a position where you needed to go to a supposed “expert” on a particular topic, how much do you think that they would be likely to know about the topic?

Please indicate, on a scale from 0 to 100, how much you think a well-regarded expert (that is, someone who is considered by many to be an “expert” on the topic) is likely to know about each topic.

0 = Most purported experts know nothing at all about the topic

100 = Most purported experts know all there is to know about the topic

0 10 20 30 40 50 60 70 80 90 100

The physical, chemical, and biological properties of food



Bees and honey farming



Astrophysics



History



3 – predictive outcome generalization

Potential risks with being overconfident:

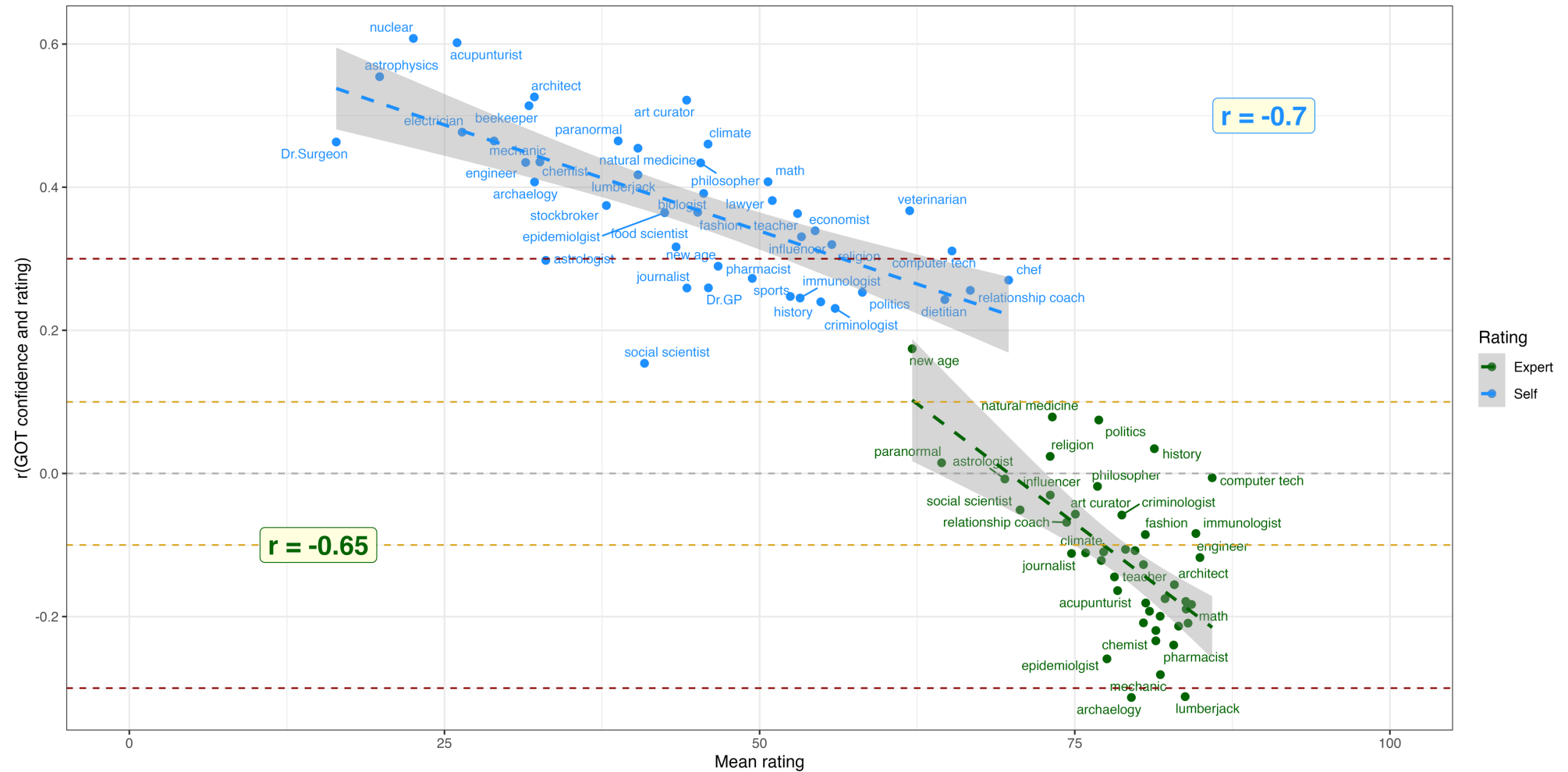
- Rely on own opinions *too* much (self-rating)
- Rely on opinions of experts *too* little (expert-rating)
- Potentially both (difference score)

3 – predictive outcome generalization

S3: GOT -> Trust in experts

	Self-ratings *Controlling for expert ratings	Expert-ratings *Controlling for self ratings	Difference score Expert – self rating
GOT	$(\beta = .56, p < .001)$	$(\beta = -.12, p = .133)$	$(\beta = -.45, p < .001)$
AOT	$(\beta = -.15, p = .021)$	$(\beta = .11, p = .140)$	$(\beta = .16, p = .012)$
IH	$(\beta = -.04, p = .514)$	$(\beta = .28, p < .001)$	$(\beta = .18, p = .003)$
Overclaiming	$(\beta = .06, p = .262)$	$(\beta = -.09, p = .116)$	$(\beta = -.09, p = .082)$

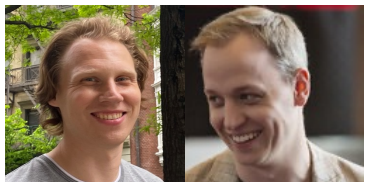
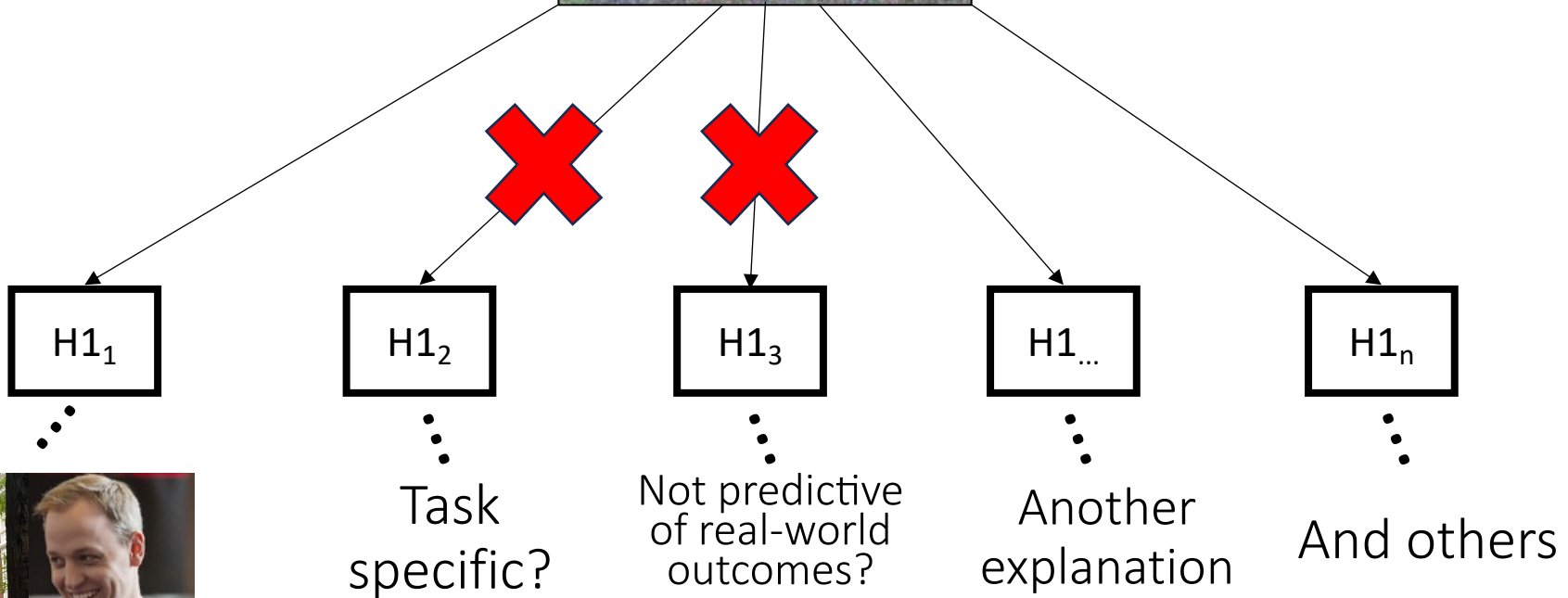
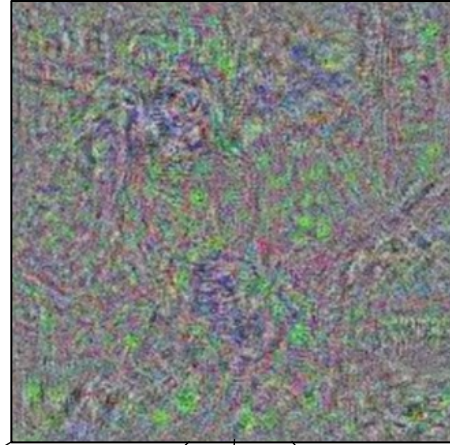
3 – predictive outcome generalization



Study 3: Summary

- GOT was a significant predictor of self-ratings and difference score
- Item-analysis pattern of findings
 - Higher overconfidence had higher self-ratings for *most* difficult domains
 - Higher overconfidence had *lower* ratings for most trusted professions

Finding



“Obvs proves OC is a trait”

Overall Summary

- There is evidence for a general overconfidence
 - Measurement confounds hindered ability to detect it
- The GOT provides a new and novel measurement approach
 - 1 - Evidence supporting its validity and reliability
 - 2 - GOT approach is generalizable to other tasks
 - 3 - Predictive of overconfidence in real life (e.g., beliefs, updating, and trust in experts)

Future directions

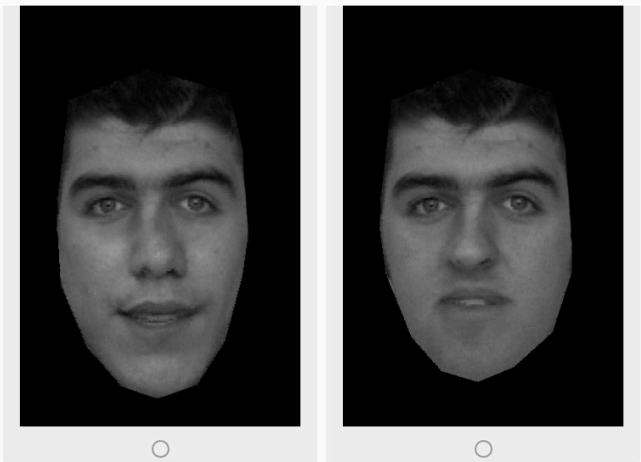
- What leads to differences in general overconfidence?
 - Representational and revision accounts



Future directions

- How pliable is one's general overconfidence?

Which face is real?

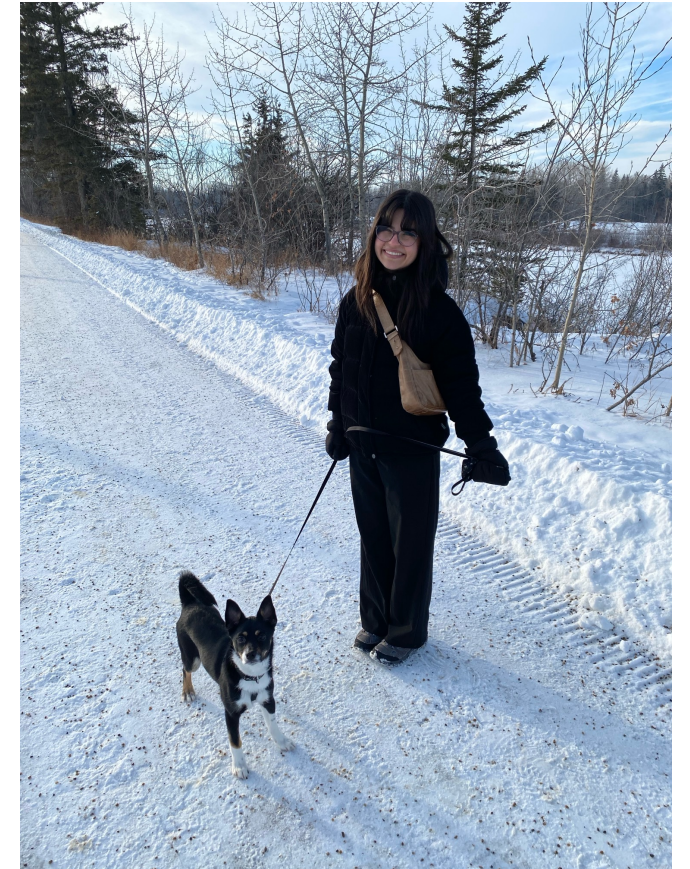


How **confident** are you that you are correct?

not at all confident highly confident

0 10 20 30 40 50 60 70 80 90 100

A horizontal slider bar with a blue circle marker positioned at approximately 35.



Future directions

- Trait / state interaction
 - Role of task difficulty, ambiguity, and familiarity

Psychonomic Bulletin & Review
1997, 4 (3), 387–392

The domain specificity and generality of overconfidence: Individual differences in performance estimation bias

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and

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One hundred twenty-three college students performed a knowledge assessment task and a game of motor skill in which they had to predict their performance before each block of trials. There was

Questions?

Scan to access GOT Resources

